

General Pathology

Final coaching
NOV-DEC 2026 DLE

Dr. Nikko D. Liberato

① McCune Albright ② Jaffe

1. A severe polyostotic fibrous dysplasia of bone involving nearly the entire skeletal system including the jaws with "cafe-au-lait" spots of the skin and precocious puberty in affected females is

↳ many bones affected

"early"

* weaker and less dense bone

A. Albright's syndrome

Rx: Ground-glass

B. Weil's - Leptospirosis

C. Jaffe's - normal growth and development

D. Costen's - TMD related

2. Peutz-Jegher's Syndrome has a tendency to develop malignant transformation in the

"HIPS"

Oral Manifestation: melanin pigmentation of the lips

A. Liver

B. Colon

C. Oral Mucosa

D. Pharynx

Gardner's Syndrome
"king of multiples"
- multiple polyposis
- multiple supernumerary

3. Tissue whose cells are continuously proliferating and can easily regenerate after injury:

A. Labile Tissues ie. Epidermis, Bone Marrow

B. Stable Tissues - moderate, can multiply if initiated/stimulated
ie. Liver, kidney, Pancreas

C. Permanent Tissues - cannot divide
ie. Sk, cardiac muscle, neurons

D. None of the above

4. Heart-failure cells are

A. Aschoff's giant cells

↳ multinucleated macrophages

B. Anitschkow's myocytes

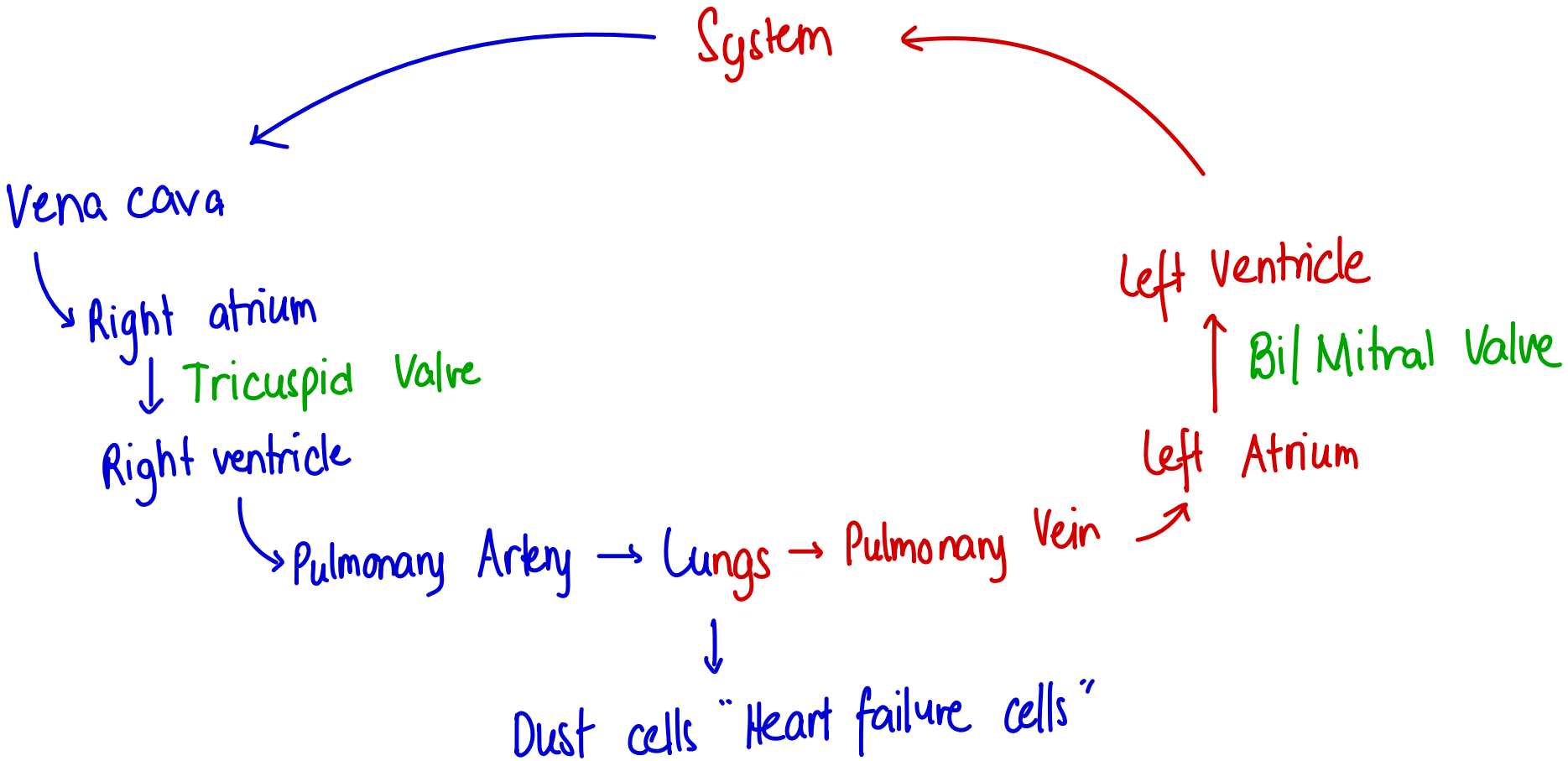
↳ large macrophage "caterpillar cells"

} Pathognomonic of RF

C. Hypertrophic Myocardial fibers

D. Hemosiderin-laden macrophages in alveoli

Left sided failure
↓
Blood pools in the lungs
↓
Leakage of RBC
↓
Alveolar macrophages "Dust cells" eats the RBC
↓
Breaking down of hemoglobin to hemosiderin


System
Vena cava
Right atrium
Tricuspid Valve
Right ventricle
Pulmonary Arkery
Lungs
Pulmonary Vein
Left Atrium
Bi/Mitral Valve
Left Ventricle
Dust cells "Heart failure cells"

multiple OKC

5. Bifid ribs, multiple radiolucent lesions of the maxilla and the mandible, multiple skin lesions and calcification of the falx cerebri are characteristic of

↳ nevus

feature during CT scan

A. Horner's syndrome - "drooping of the eyelids"

B. Gardner's syndrome

C. Sturge-Weber syndrome - Vascular malformation
- Encephalotrigeminal Angiomatosis "port-wine stain"

D. Peutz-Jeghers syndrome

E. basal cell nevus syndrome. Gorlin Goltz | NBCCS

"Gorlin Sign" - Ehler Danlos (tongue to the tip of the nose)

* Question w/ contradicting choices

6. Cause of edema can be described as

↳ fluid in interstitial spaces

A. Decreased plasma protein concentration fluid goes to higher concentration (↑ C.)

B. Increased lymph flow - would remove excess interstitial fluid
- reduce edema

C. Increased ECF volume - a result or a contributing factor of edema

↱ fluid goes into circulation

D. Increased plasma protein concentration

↓ Plasma Protein → ↓ Lymphatic Drainage

↑ Hydrostatic Pressure → ↑ ECF Volume

7. Which of the following potential carcinogens is most like to produce free radicals?

↳ atoms w/ unpaired electrons

A. An alkylating agent - anti-cancer drugs (ie. cyclophosphamide)
"paradox of chemotherapy" - Damaged DNA from cells may survive and mutate

B. Ultraviolet light

C. A virus ie. Hepa C Virus (Flavivirus)

D. Ionizing radiation - free radicals interact to DNA

8. In hypovolemic shock

↳ ↓ fluid volume

A. The central venous pressure is high — Low BV = Low BP

B. The extremities are pale, cold and sweating → sympathetic stimulation of sweat glands

↳ ↓ friction in the BV

C. There is always a site of bleeding → could also be due to loss of other body fluids

↓

loss of circulating volume

ie. Dehydration, Diarrhea, burns, 3rd spacing (cavities)

D. Urine output is ~~unaffected~~

↳ renal function regulating hemostasis

3 main types

① Hypovolemic
- most common
- loss of blood or plasma

② Cardiogenic
- due to conditions affecting CO (ie. ischemia, M.I.)

③ Distributive

ⓐ Anaphylactic
- allergic reaction

ⓑ Neurogenic
- nerve injury affecting sympathetic tone

ⓒ Septic
- causes an immune response (TSS)
- impaired tissue perfusion

Stages
① Initial - ↓ CO
② Compensatory - ↑ HR
③ Progressive - due to failure of stage 2
- anaerobic metabolism
↓
Lactic acid
④ Refractory
↳ irreversible
↳ leads to organ failure

9. Non-painful lesions known as chancre sores occur in which stage of syphilis?

"Lues"

A. Primary - 2 - 6 weeks

B. Secondary
- Maculopapular rash, Condyloma latta (after 6 weeks)
- highly infective stage

C. Latent

* Condyloma Acuminatum (HPV -> genital warts)

D. Tertiary - Gumma (palate / tongue)

↳ usually in venous circulation

10. Sudden obstruction of some part of the vascular system through the impaction of undissolved material carried there by the blood current is known as

↳ clot, air, fat → Embolus

* Virchow's Triad (Vir CHEaw)
  Changes in blood flow
  Hypercoaguability of blood
  Endothelial Injury

A. Embolism

B. Infection

C. Infarction - sequel to embolism

D. Thrombosis - solid mass of clotted blood (Thrombus)
- agonal, mural, red, white, fibrin

E. Organization

11. Staging of malignancies is of value in determining the

A. Etiology

B. Prognosis

C. Diagnosis

D. Pathogenesis

E. Differentiation

12. The "jigsaw puzzle" appearance of bone is peculiar to:

referring to histologic, not radiographic feature
- due to continuous deposition, resorption

A. Osteitis deformans — Paget's — Rx: Cotton wool

B. Fibrous Dysplasia — Rx: Ground glass

C. Ossifying fibroma — Rx: Well Circumscribed — RL/RP (mixed) — Loc: Mandibular Anteriors

D. Osteogenic Sarcoma — Rx: Sunburst

13. Type of necrosis wherein the dead cell undergoes extensive autolysis, caused by the release of lysosomal hydrolases:

↳ enzymes → tissue becomes soft, liquid

A. Coagulative necrosis → due to ischemia/hypoxia (ie heart, lungs)

B. Liquefactive necrosis ie. (brains, lungs)

C. Caseous necrosis → cheeselike material (ie. TB, Histoplasmosis)

D. Fat necrosis → organs appear white and chalky (ie. Pancreas, Breast)

Gangrenous - "Putrefaction"

Fibrinoid - RHD, autoimmune diseases (affects arterial walls)

14. A disease characterized by small, papulovesicular greyish-white lesions commonly found at the soft palate and pharynx

a. Summer Illness (Herpangina)

b. Hand-Foot-Mouth disease - Loc: Buccal Mucosa, Hard Palate

c. Herpes Simplex

d. German Measles Rubella

15. A patient has an increased number of columnar cells in the lower esophagus. He has which of the following change?

↳ not native to the site
(squamous → columnar)

A. Dysplasia - disordered arrangement

B. Anaplasia - loss of differentiation (hallmark of malignancy)

C. Metaplasia - adapted according to function

* Barrett's Esophagus
→ premalignant condition

D. Normal histology

16. An excess of fluid in the interstitial tissue or serous cavities:

A. Edema (can be either transudate/exudate) more of a clinical finding

B. Transudate

C. Exudate

D. Pus

Transudate	Exudate
- due to ↑ hydrostatic pressure	- due to inflammation ↑ vascular permeability
- due to ↓ blood oncotic pressure	- cloudy, ↑ protein content
- clear (mainly fluids)	- ↑ specific gravity

17. Dangers of biopsy include the following EXCEPT:

A. Spread of infection
↳ breaks in epithelial barrier can introduce bacteria and spread infection
"tumor seeding"

B. Haemorrhage

C. Operative trauma ie. Nerve Injury
Pain, Swelling

D. None of the above

Types:
① Incisional - only a portion of the lesion
② Excisional - "Exit"
- entire lesion
③ Punch - small cylindrical sample
④ Exfoliative - collection of surface epithelial cells
(smear, brush biopsy)

18. Rubeola commonly produces an exanthem of the buccal mucosa designated as

measles — ↳ widespread rash

A. Negri bodies - Inclusion Bodies (Histopath)

Rabies

B. Fordyce's spots - enlarged sweat glands

C. Paschen bodies - small pox

D. Koplik spots

E. Guarnieri bodies - small pox / variola virus

19. Caseous necrosis is associated with

"cheese-like" common in granulomatous diseases

A. Pulmonary tuberculosis

B. Soft tissue abscesses

C. Sore throat

D. Thrombophlebitis

↳ inflammation of a vein due to a thrombus

20. Choose a tumor suppressor gene | anti - oncogene
↳ regulate cell division, growth, proliferation

A. myc

B. fos } Oncogenes

C. ras - responsible for CAUSING majority of all known cancers

D. Rb - Anti - retinoblastoma

p53 - "guardian of the genome"
- responsible for preventing majority of all known cancers

21. Ascites is the accumulation of excess fluid in the ~~pleural~~ peritoneal cavity ✗ In 85% of the cases, ascites is caused by cirrhosis ✓

* Pleural effusion → general term (hydro, hemo, pneumo)

a. Both statements are true

b. Both statements are false

c. The first statement is true, the second statement is false

d. The first statement is false, the second statement is true

Others
- pericardial effusion
- hydrocephalus

Cirrhosis → portal hypertension ↓ ↑ pressure pushes fluid out of the hepatic vessels

22. A subdivision of pathology that deals with the study of structural changes in diseases:

A. Etiology

B. Pathogenesis - how a disease develops

C. Morphology

D. Clinical Significance

extra finger / toe

23. An infant with cleft lip, palate, polydactyly, microcephaly with holoprosencephaly, and ectodermal scalp defect is suffering from

holo = whole
prosencephalon = forebrain
- forebrain fails to properly divide into 2 hemispheres

A. Trisomy 21

Down
- large tongue

B. Trisomy 18

Edwards
- micrognathia, short neck, overlapping fingers

C. Trisomy 13

Patau

D. Turner syndrome

XØ (only females) → ovarian failure, webbed neck
aortic abnormalities, short stature
- barr bodies : X chromosome is inactive
- sex linked; NOT Autosomal

24. Right sided heart failure results in:

A. Pulmonary edema - due to left sided failure
- leads to right sided failure

B. Peripheral edema

C. Lymphedema

D. Cirrhosis of the liver

25. The hereditary transmission of Peutz-Jegher's Syndrome is

A. Autosomal dominant

- 1 mutated gene
- affects males and females equally
- * If a parent has the disease = 50% passing it to each child

ie. Marfan, Achondroplasia

B. Autosomal Recessive

- parents can be healthy carriers
- 25% chance of disease, 25% unaffected, 50% carrier

ie. CF, SCD

Tay Sachs, PKU

C. Sex-Linked dominant

D. Sex linked recessive

} x or Y

more common in males because of only one X chr (no other X chr to compensate)

ie. Hemophilia A

females are carriers

E. Not genetically transmitted

26. Biopsy that is aimed at the complete surgical removal of the lesion for diagnostic therapeutic purposes:

A. Incisional biopsy

B. Excisional biopsy

C. Punch biopsy

D. Exfoliative cytology

* Some references has XO on letter A and C

27. Females with Turner Syndrome have the following sex chromosomes:

A. XO

B. XXX - mild to no symptoms

C. XXY - Klinefelter → Hypogonadism in males
- testicular atrophy
- Infertility
- Gynecomastia

D. XYY - Jacob's Syndrome (mental problems
- learning
- behavioral)

28. An autoclave sterilizes dental instruments by causing which of the following?

121° C @ 15 psi for 15 - 20 mins

A. Coagulation of proteins

B. Denaturing of proteins

C. Precipitation of nucleic acids

D. Disruption of cell membranes

} Alcohol, Detergents

E. Dissolution of lipids

29. **Metastatic calcification** is seen in

↳ Ca^{+} deposits in normal tissues (kidney, lungs, gastric mucosa, BV)

Due to hypercalcemia

~~A~~. Vitamin A deficiency

~~B~~. Vitamin D deficiency → causes hypocalcemia

C. Hyperparathyroidism ↑ PTH = ↑ bone resorption and is replaced by fibrous tissue, cyst

D. Hypothyroidism - most common cause: Hashimoto's Thyroiditis due to iodine deficiency

- cold intolerant, cool skin, fatigue

30. Tumor composed of multiple tissues in which there may be representatives of all three embryonal layers is a(n)

A. Teratoma — may contain hair, teeth, bone, skin

B. Adenoma — "gland" - Mixed tumor (epithelial + mesenchymal tissue)

C. Carcinoma - ectodermal / endodermal

D. Sarcoma - mesoderm

E. Hamartoma - local; mixture of cells and tissues

31. Which of the following is common in all forms of shock?

A. Sepsis Septic Shock

B. Hypovolemia Hypovolemic Shock

C. Vasoconstriction Vasodilation present in Anaphylactic Shock

D. Impaired tissue perfusion due to low circulating BV

32. Which of the following tumors has the best prognosis for patient survival?

A. Hepatoma - no s/s
- late detection

B. Melanoma of the gingival - high vascularity

C. Osteogenic sarcoma of the femur - highly aggressive, early metastasis

D. Adenocarcinoma of the gallbladder - high vascularity, late s/s

E. Squamous cell carcinoma of the lip - early detection, localized, superficial growth, easy accessibility for treatment

Tx: Full thickness resection

33. Following injury, which tissue restores functional capacity by hypertrophy

↳ cellular adaptation ↑ cell size

- minor injury : 100% recovery
- major injury : leads to cirrhosis

A. Liver (hyperplasia)

B. Adrenal Cortex

C. Cardiac Muscle

D. Peripheral nerve

E. Skin of the trunk

34. T1 N1 M0 lesion is

A. Stage I T_1 , T_2 } No nodal involvement, No metastasis

B. Stage II T_3 , T_4 }

C. Stage III T_1 - T_4 . w/ nodal involvement, w/o metastasis

D. Stage IV w/ metastasis

35. "Hair-on-end" appearance in a skull roentgenogram is seen in

"Buzz - cut" appearance

A. Fibrous Dysplasia - Ground glass

B. Thalassemia

C. Garre's Osteomyelitis - Onion Skin (Type of Chronic Osteomyelitis)

D. Paget's disease - Cotton wool appearance

Familiarize : Radiographic Appearances
Histopathologic features

36. Characteristics of plasma cell myeloma EXCEPT

Multiple Myeloma

A. Malignant condition

B. Produces ~~functional~~ immunoglobulins

non functional

C. Bence-Jones proteins are found in the urine

D. Radiographic findings include "punched-out" radiolucencies in bones

E. Usually arises in the bone marrow

Pathologic fractures due to resorption (spine, femur)

37. Bence Jones proteins are associated with

A. Hodgkin's lymphoma

↳ Hx feature: Reed-Sternberg Cells "Owl's Eye"

B. Burkitt's lymphoma

↳ associated w/ EBV Hx feature: "Starry Sky"

C. Multiple myeloma

D. Infectious mononucleosis "Kissing Disease

38. It is the most common malignant tumor of childhood and infancy

A. Neuroblastomas - Forms in the adrenal medulla
↳ because it arises from neural crest cells

B. Pheochromocytoma - tumor of Adrenal made up of chromaffin cells : ↑ E NE
↑ BP
Sympathetic Stimulation

C. Neurofibroma - benign tumor of Peripheral Nerves
(Neurofibromatosis type I) VRD

D. Osteoid Osteoma
Common Site : Cortex of Femur, Tibia

(non-sex chromosomes)

39. Which of the following autosomal chromosomal abnormalities occur most frequently?

22 pairs

A. Edward syndrome Trisomy 18 — 1 in 8,000 births
- micrognathia, short neck, overlapping fingers

B. Patau syndrome Trisomy 13 — 1 in 15,000 births
- cleft palate, lip, polydactyly

C. Klinefelter syndrome → sex linked

D. Down syndrome Trisomy 21 — 1 in 700 births

E. Turner syndrome → sex linked

40. Which of the following is FALSE concerning necrotic changes?

A. Ischemia is seen with coagulative necrosis. Loss of blood supply

B. Fatty acids are released with liquefactive necrosis.
↳ high lipid content of the brain → breakdown of these lipids * different mechanism w/ fat necrosis (saponification)

C. Myocardial infarcts cause ~~liquefactive~~ coagulative necrosis.

D. Caseous necrosis is seen in tuberculosis infection.

E. Granulomas are considered as being characteristic of caseous necrosis.

41. Which of the following definitions is ~~NOT correct~~ regarding classical signs of acute inflammation?

False

A. Calor due to decreased tissue perfusion (heat due to ↑ blood flow)

↳ should be increased

B. Tumor due to tissue edema - swelling

C. Rubor due in part to vasodilation - redness

D. Function laesa due to pain and swelling - loss of function

E. Dolor due to inflammatory mediators - pain

42. The term carcinoma refers to a:

A. Benign Mesenchymal tumor Fibroma

B. Benign Epithelial Tumor Papilloma, Adenoma

C. Malignant Mesenchymal Tumor - Sarcoma (CT: bone, muscle, fat)

D. Malignant Epithelial tumor

E. None of the above

43. A man was found unconscious with zero vital signs. After several attempts of CPR he was pronounced deceased. Autopsy revealed that he suffered from massive myocardial infarction. Which category of shock did this man suffer from?

↓
due to lack of
blood supply to the heart
↓
↓ cardiac output
↓
Circulatory Disturbance

A. Septic

B. Cardiogenic - ↓ cardiac output

C. Anaphylactic - allergic reaction

D. Hypovolemic - ↓ fluid volume

44. Fibrosis plugin over epithelial regeneration resulting in slower healing, with more inflammation and granulation tissue formation, and more scarring:

A. First Intention → clean, approximated edges (w/ sutures) → minimal tissue loss, granulation, scarring → faster healing

B. Second Intention → large tissue loss, wound not approximated, slower healing

C. Third Intention → wound kept open to control infection, then closed using first intention

D. Fourth Intention

45. Cystic fibrosis is a hereditary disorder that results from defective

↳ autosomal recessive , fatal

A. Collagen

B. Lysosomal enzymes · LSD

C. Chloride channels - encoded by CFTR gene

* sweat test - ↓ Cl reabsorption

D. Fibrillin

E. Myelin

Airways and lungs
- thick and sticky mucus (↓ Cl secretion)
↓
Chronic infections
↓
respiratory failure

↓
↑ Na, H_2O reabsorption

Lysosomal Storage Diseases
- causes build-up of toxic material in our cells due to defective lysosomal enzymes → organ damage

① Tay Sachs
enzyme: Hexosaminidase A
"cherry-red" spot formation in retina causing blindness

② Niemann Pick

③ Gaucher

④ Fabry

Glycogen Storage Diseases (GSD)

- Enzyme breaking down glycogen is defective
  - ↓
  - glycogen builds up → liver damage
  - ↓
  - Low blood glucose/sugar

① Von Gierke
② Pompe's
③ Hers

46. An infant with a diagnosis of osteopetrosis has dysfunctional

"petrous" - hard/dense

A. Chondrocytes

B. Osteoblasts

C. Osteoclasts

D. Fibroblasts

E. Lymphocytes

Albers - Schonberg

Rx: GG

↑ sclerotic bone bass

↑ hardness = ↓ tensile strength

"Marble bone disease"

Hardest part of the skull

↳ Petrous part of the temporal bone

47. The etiology of acute diffuse glomerulonephritis seems to be

Hypersensitivity type III : Immune Complex Sensitivity

A. Circulatory deficiency associated with prolonged shock

B. ~~Bacteremia~~ with localization of organisms in kidney tissue

↳ due to Ab - Ag Complexes, not bacteria

C. Injury of glomeruli by exogenous inorganic toxins

D. Degenerative changes induced by scleroticalterations of blood vessels

E. Allergic reaction of glomerular and vascular tissue to beta-hemolytic streptococcal products

Ag - Ab complexes
↓
Attracts anti-inflammatory cells
↓
Damage glomerular Capillary wall
↓
Hematuria

COPD

48. It refers to the irreversible enlargement of the airspaces distal to the terminal bronchioles, accompanied by destruction of their walls without obvious fibrosis.

A. Chronic bronchitis - mucus hypersecretion
- may predispose to bronchogenic Ca

"blue bloaters"
3 months of productive cough

B. Bronchiectasis - permanent dilation of alveolar wall (due to chronic infection)
- w/ fibrosis because of repeated inflammation and repair

C. Cystic Fibrosis

D. Emphysema → enzymatic destruction → no fibrosis because the injury is destructive (no fibroblast activation)

"phro" - protein

49. Nephrotic Syndrome, a glomerular disease is characterized by all of the following EXCEPT one. Which one is the EXCEPTION

Triad
- heavy proteinuria > 3.5g/day
- hypoalbuminemia
- edema

A. Heavy Proteinuria

B. Lipiduria - excess gets secreted

C. Hyperalbuminemia

D. Hyperlipidemia
- to compensate for ↓ blood oncotic pressures
- uses lipids as solutes

E. Severe Edema

Nephritic → hematuria, proteinuria
→ post infection w/ strains of Group A, beta hemolytic strep

50. Addison's disease is an endocrine disturbance caused by hypofunction of the

↳ Adrenal Insufficiency

* Negative feedback system

A. Adrenal cortex "GFR"

B. Adrenal medulla Catecholamines → NE, E, D

C. Anterior lobe of the pituitary ↑ ACTH to stimulate adrenal cortex

Adenohypophysis

* TSH, FSH, GH, Prolactin

D. Posterior lobe of the pituitary

Neurohypophysis

* ADH, Oxytocin

51. In chronic granulomatous inflammation, which of the following groups of cells are characteristic?

A. Neutrophils, mast cells and basophils → acute

B. Neutrophils, eosinophils and histiocytes
chronic

C. Lymphocytes, plasma cells, and macrophages

D. Basophils, eosinophils, and Reed-Sternberg cells → form of histocyte (macrophage)
Hodgkin's Lymphoma

E. Histiocytes, Reed-Sternberg cells, and mast cells

Acute	Chronic
- Rapid - Short - Single Stimulus - Exudation - Congestion - Transudation	- Slow - Long - Persistent Stimulus - Proliferation

52. Benign tumors are usually:

"Differentiation"
- how much tumor cells resemble original tissue

A. Well-differentiated
- mimic normal tissue, slow growth
- encapsulated (well-circumscribed)

B. Moderately-differentiated

C. Poorly-differentiated

D. Non-differentiated (Anaplasia)
Hallmark of malignancy

Malignant Cells
- Anaplasia
- Pleomorphism (nucleus)
  - hallmark of anaplasia
- Aneuploidy
- high nucleus to cytoplasm ratio

53. Lack of development of an organ or tissue, which can also refer to lack of production of cells from an organ or tissue:

A. Aplasia

B. Atrophy – ↓ in size due to disuse, pressure, loss of innervation, decreased nourishment

C. Hyperplasia – ↑ in number of cells

D. Hypertrophy – ↑ size of cells

E. Metaplasia – Replacement of one cell type w/ another

↱ ↑ PTH

54. Osteoporosis, metastatic calcifications, renal stones, increase serum calcium level are manifestations of:

↑ resorption activities

A. Hyperadrenalism "Cushing's"

Hypoadrenalism = Addison's
Hypocortisolism = Cushing's

B. Hyperparathyroidism

C. Hypoparathyroidism - leading to tetany (Chvostek sign, trousseau's)

D. Hyperthyroidism - inc. metabolism, warm body, heat intolerant, moist skin

55. Which of the following is a SIGNIFICANT effect of pheochromocytoma?

↳ tumor of adrenal gland consisting of chromaffin cells
↳ sympathetic stimulation

A. Myxedema

B. Acromegaly

C. Glycosuria

D. Hypertension

56. Which of the following describes the hereditary transmission of hemophilia A.

Factor VIII deficiency

A. Autosomal Dominant

- affects male and female equally
- only one mutated gene needed, appear in every generation

B. Autosomal Recessive

- males and females equally
- 2 mutated genes required; parents are usually carriers, skips generations

C. X-linked

D. It is not genetically transmitted

57. Which of the following cancers has the highest incidence in men?

A. Colorectal
- 3rd highest in incidence
- 3rd highest in cause of death

B. Urinary tract

C. Prostate
- overtook lung cancer in incidence due to widespread screening (PSA testing); 2nd highest cause of death

D. Lung
- 2nd highest incidence
- highest in cause of death

E. Leukemia

58. Pernicious Anemia is associated with the follow condition:

A. Gastric Hyperacidity

B. Microcytic Anemia

C. Glossitis

D. Iron-Deficiency

Anemia → RBC Disorder (↓oxygen carrying capacity)

- iron deficiency - not enough hemoglobin to carry O_2 effectively
- Megablastic - large RBC due to Vit B12 and folate deficiency
- Pernicious - autoimmune (B12 malabsorption)
  - lack of intrinsic factor
    - ↳ released by parietal (oxyntic)
    - ↳ B12 absorbed in terminal ileum
      - * Schilling test
- Hemolytic - "SETH"

ⓐ Sickle cell - obstruction of capillaries
- severe pain

ⓑ E. fetalis - Rh incompatibility

ⓒ Thalassemia - beta (Cooley's anemia)
Rx: Hair-on-end

ⓓ Hereditary Spherocytosis

- Aplastic Anemia
  - abnormal stem cells
  - suppression of bone marrow

} Pancytopenia
↓ all blood cells

59. DiGeorge syndrome is characterized by a deficiency of

↳ chromosomal abberation of Chr 22

A. B lymphocytes

B. T lymphocytes

C.Both B and T cells

D. antibodies

C ranial defects
A bnormal facial features
T cell deficiency
C left Palate
H ypocalcemia

Defect of 3rd and 4th Branchial Arches

↓

Underdeveloped Thymus
Underdeveloped PTH gland

↓

↓↓ T-lymphocyte maturation

Hypocalcemia causing Tetany

Chr 22 Deletion

60. The commonest site of thrombosis is

Take into consideration Virchow's Triad

A. Arteries - less frequently (↑ pressure)
- arteries supplying the heart ⟶ causes M.I., stroke

B. Veins - esp deep veins of the legs
- due to slow blood flow, prone to stasis

C. Capillaries - not typical sites

D. Heart - mural thrombi (after M.I.)
- less common

61. Shock is a circulatory disturbance characterized by

A. ~~Increased~~ blood pressure

B. Decreased volume of circulating blood

C. Elevated body temperature

D. Decreased volume of interstitial fluid

62. Which of the following statements is INCORRECT?

A. Plasma cells come from T-lymphocytes

↳ B - lymphocytes

B. Neutrophils can increase in chronic bacterial infections

C. Prostaglandins cause pain

D. Multinucleated giant cells are fusion of many macrophages

63. When a thrombotic embolus originates in a femoral vein, it usually becomes arrested in the

↳ dislodged thrombi

A. Right heart - too wide

B. Renal circulation

C. Portal circulation

D. Hepatic circulation

} From internal organs

Pulmonary Embolism

↓

Shortness of breath
Chest pain
Hypoxia

E. Pulmonary circulation

"Gustation" - sense of taste

64. **Gustatory sweating** is the chief symptom in

A. Bell's palsy - CN VII - due to deposition of LA (transient), Ramsay Hunt, Tumor

B. Meniere's disease - Vertigo (CN VIII)

C. Orolingual paresthesia - tickling / prickling sensation (tongue area)

D. Auriculotemporal syndrome Frey Syndrome - common after parotidectomy
↓
severed nerves of parotid
innervates local sweat glands
↓
Instead of Salivating = Sweating
Loc. of sweating
- temple, back of the ear, cheeks

E. Glossopharyngeal neuralgia
↳ burning sensation
@ throat area

65. Marie Strumpell Disease affects the:

"Ankylosing Spondylitis"

A. Knee joint

B. Spine

C. Pelvis

D. Heart

66. Nephrolithiasis is most likely to be associated with which of the following conditions?
kidney stone condition

A. Hyperparathyroidism ↑ PTH → ↑ Resorption → ↑ serum calcium

B. Myxedema → severe hypothyroidism in adults
Cretinism → in children

C. Pyelonephritis
pus kidney inflammation

D. Wilson's disease → copper accumulation in brain, liver, etc.

E. Thrombocytopenia - autoimmune disease targetting platelets
platelet decreased

67. Karyotyping can be used to diagnose which of the following diseases?

↳ test to examine chromosomes

A. Klinefelter Syndrome

B. Multiple Myeloma

C. Niemann-Pick disease

D. Pemphigus

E. Peutz-Jeghers

68. Anasarca means

A. Abnormal inflammatory process

B. Severe generalized swelling

C. Absence of proliferation of vessels following inflammation

D. Presence of pus

69. A chromosomal aberration in chromosome number 22 is associated with:

↳ Also: DiGeorge Syndrome

A. Down's Syndrome — Chr 21

B. Chronic Myelogenous Leukemia — Philadelphia Chromosome: translocation b/w chr 22 ⇆ 9

C. Aplastic Anemia
- abnormal stem cells
- suppression of bone marrow

D. Turner's Syndrome — – X∅

Leukemia

- Lymphoid - lymphoblastic (immature) / lymphocytic (mature) → B cell, T-cell, NK cell
- Myeloid - myelogenous → erythrocytes, granulocytes, monocytes

① ALL - most common childhood leukemia

② AML - common in 60 and above

③ CLL - good prognosis

④ CML - Translocation of portions b/w Chr 22:9 (Philadelphia Chromosome)

70. Incidence of most common malignant tumors in women is

A. Breast

B. Lung - most fatal in both men and women

C. Cervix

D. Ovary

71. Venous emboli are most often lodged in

A. Intestines

B. Kidneys

C. Lungs

D. Heart

72. Which of the following is a histologic hallmark of malignancy?

A. Dysplasia

B. Anaplasia

C. Metaplasia

D. Desmoplasia - formation of fibrous connective tissue around a malignant tumor

73. Type of necrosis seen in hemorrhagic infarction of the lung is

due to ischemia

A. Caseous necrosis

B. Coagulation Necrosis

C. Fibrinoid necrosis

D. Gangrenous necrosis

→ abnormal accumulation of fluid in interstitial/extracellular spaces or body cavities

74. Edema may be caused by any of the following, except

A. An ~~increase~~ in the plasma protein concentration

decrease albumin

(osmotic pressure)

fluid moves to a higher concentration

B. An increase in the capillary hydrostatic pressure

C. An increase in the capillary permeability

} Allows more fluid to accumulate in I.C

D. Lymphatic obstruction

↳ fluid cannot drain; stuck in IC

75. Which of the following is true about nephrotic and nephritic syndrome?

A. Nephrotic syndrome involves loss of a lot of protein, whereas nephritic syndrome involves the loss of a lot of blood

B. Nephrotic syndrome is more commonly associated with edema than nephritic syndrome due to ↓ proteins in the circulation

C. Hematuria is eminent in nephritic syndrome

D. All of the above are true

76. Cellular atrophy is caused by the following EXCEPT:

↳ ↓ cell size (cell wasting)

A. Loss of blood supply of innervations - pressure | nerve injury

B. Loss of endocrine factors - endometrial atrophy after menopause

C. Decrease in the workload

D. Aging and chronic illness ie. Brain, Skeletal muscle

E. None of the Above

77. Malignant neoplasms show the following features except

A. Disorganizes cell structure - anaplasia
- pleomorphism

B. Encapsulation Characteristic of Benign tumor
↳ slow growth, well-defined borders

C. Invasion of blood vessels (metastasis)

D. Rapid, erratic growth

* Pathognomonic

78. Cullen's sign and Turner's sign are usually associated with which of the following conditions?

↳ flank ecchymosis

↳ Bluish discoloration around umbilical area

A. Acute Pancreatitis

Dunphy - pain during coughing
Rovsing - pain on LRQ if you press LLQ
Obturator - pain during hip flexion
Psoas - pain during thigh extension

B. Appendicitis

C. Gastroenteritis

D. Oophoritis - inflammation of the ovary

* Levine Sign (angina pectoris)

79. All of the following are typically associated with the loss of 40% of the circulating blood volume, except

A. A decrease in the blood pressure ↓ BV ⟶ ↓ BP

B. A decrease in the central venous pressure

C. A ~~decrease~~ in the heart rate - should be increased due to compensation

D. A decrease in the urine output - fluid preservation

80. Which of the following is a feature of necrosis?

↳ irreversible

A. Reduced cell size

B. Intact cellular contents

C. Intact plasma membrane

D. Karyolysis

81. The host tissue response in acute inflammation is all, except

A. Exudative - due to ↑ vascular permeability

B. Necrotizing - necrotic cells due to activation of WBC

C. Granulomatous - chronic (sign of tissue repair)

D. Cytopathic - cell lysis, common in viral infections

Granulation tissue
- new capillaries
- macrophage
- fibroblast

} embedded in a matrix

→ also present in Patau

82. Polydactyly, craniosynostosis, late closure of fontanelles is a feature of:

↳ early fusion of sutures causing cranial deformity

A. Apert's syndrome

B. Crouzon's syndrome

} Almost same features except Crouzon does not have polydactyly

C. Pierre Robin syndrome

D. Down Syndrome

83. The following may be transmitted from mother to fetus except one. Which is it?

Teratogens "cheap torch"

- Chx pox
- Hepa B | HIV
- Enterovirus
- AIDS
- Parvo
- Toxoplasmais
- Others (Group B strep)
- Rubella
- Cytomegalovirus
- HSV

CD4 count

HIV	AIDS
>200 - <500 cells/mm³	<200 cells/mm³

A. Hepatitis B

B. HIV

C. Rubeola

D. HSV

84. Plummer's Disease is described as, except:

most common : Graves

A. Second most common cause of hyperthyroidism

B. Exophthalmos is always present

C. Toxic multinodular goiter - other name of Plummers

D. Increases / Worsens with age - almost all diseases worsens with age

85. Hepatitis C virus infection is MOST often transmitted through

A. Sexual contact - possible but low chance

B. Oral secretions

C. Blood transfusions

D. The fecal-oral route (A, E)

E. Respiratory droplets